MINI PROJECT ABSTRACT

Project Title: Interactive Geographical Heritage Map of India

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Abstract (250–300 words):

India possesses an unparalleled cultural tapestry spanning millennia of indigenous crafts, folk dances, registered GI heritage, sacred river lifelines, and diverse regional traditions. However, conventional pedagogical mediums—such as static textbooks or physical exhibits—often fail to engage contemporary digital learners due to a lack of spatial context and multisensory interaction. To address national Problem Statement 26197 ("Innovative ideas showcasing Indian traditions"), this mini-project presents the "Interactive Geographical Heritage Map of India," an engaging web-based digital cartography platform engineered to preserve, celebrate, and digitize cultural exploration across all 37 States and Union Territories of India.

Developed utilizing pure client-side web technologies (HTML5, CSS3, and Vanilla JavaScript) powered by the open-source Leaflet.js GIS engine, the platform renders high-precision GeoJSON vector boundaries with curated state palettes alongside 35+ glowing sacred river centerlines. Users can navigate across midnight dark, high-resolution satellite topography, and warm terrain basemaps. Selecting any region launches a responsive five-tab heritage drawer detailing certified Geographical Indications (GI tags), classical dances, major festivals, iconic regional cuisines, and traditional handloom attire. Elevating the experience beyond visual cartography, the system integrates an authentic HTML5 audio subsystem delivering forced five-second acoustic snippets from field recordings, a side-by-side dual state comparison engine, and a gamified heritage trivia challenge.

This zero-dependency, static web architecture guarantees rapid rendering, zero server-side maintenance costs, and robust cross-device responsiveness without requiring bloated compilation frameworks. Ultimately, this interactive cartography application serves as an accessible educational tool that seamlessly unites ancient Indian traditions with modern front-end web engineering, fulfilling the institutional mandate to innovatively showcase and preserve India's living cultural legacy for future generations.

Keywords:
Cultural Heritage, Interactive Cartography, Leaflet.js, GeoJSON Vectors, Sacred Rivers, Multisensory UI, Geographical Indications, Indian Traditions, Front-End Web Engineering.